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Mobility of Indonesian during Early Pandemic: Insights from Mobile Positioning Data

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***Abstract*—Mobile Positioning Data (MPD) contains information on the location of the mobile phone by approximating mobile phones' location relative to fixed infrastructures (e.g., telecommunication towers that transmit signals). While the data query is technically straightforward, obtaining this dataset requires particular permission to protect customers' privacy. Additionally, the dataset has large volumes of data (i.e, up to 300GB per day), resulting in not many researchers holding this data source to analyze the mobility of people. In this work, we collaborate with one of the biggest telecommunication service providers in Indonesia to collect MPD and prepare the big data infrastructure. We thus analyze mobility patterns during the early phase of COVID-19 in 2020 using actual Mobile Positioning Data in five provinces in Java. We use three metrics, namely, the number of visits, averaged travel distance, and Origin-Destination matrix. The findings indicate that the social restriction in the corresponding provinces has reduced the average traveled distance of the people, but not their number of visits. That is, while the traveled distance has declined more than eight times compared to the baseline, the number of visits may rocket up, up to nine times. It indicates that people are still having shorter trips even though their regular activities (working, schooling, etc.) have been restricted. The data also show that during Ramadhan month, the government has a successful intervention in restricting people for mudik Lebaran, The number of visits dropped to below 30 visits during Ramadhan and only small spikes exist during *'libur lebaran'*.**

***Index Terms*—big data, visualization, mobility, pandemic, positioning**

I. Introduction

The mobility of people is a measure of movement from one place to another place of a given population in a given time frame. The mobility analysis is paramount in investigating transportation problems as it entails the analysis of movement trends, the hotspot of points of interest, and communal activity patterns. Mobility analysis is also used for improving public transportation planning, such as taxi services [1]. When a disease spreads rapidly over a large area, population mobility may also provide insight into how the disease spreads, especially for viral diseases that spread among humans. The coronavirus disease (COVID-19) is an example. In the early phase of the spread, people are totally aware that localizing an contagious area (i.e., lockdown) is the best way to hold the spread down. However, restricting people movement may raise psychological problems and economic consequences. By having information on population-level mobility, authorities may take effective policies that have minimum impacts while preventing the spreads. Additionally, mobility analysis may reveal how far the social distancing is being obeyed [2].

The "Big Tech" companies like Apple[1], Facebook[2], and Google[3] have opened aggregated datasets of their users' movement trends. While these private sector data may help review governments' policy impacts, some datasets may be biased, such as over-represent higher-income populations or skewed towards lower-income groups. Another approach is by conducting surveys (e.g., McGuckin and Fucci [3]). A travel survey explores travel behavior by asking directly to respondents that represent a population (e.g., number of trips made, the origin and destination, departure and arrival times, etc). However, this approach may suffer from sampling biases and reporting errors (e.g., unreported trips). Moreover, surveys are tedious and expensive to conduct. Consequently, surveys are organized sparsely, such as once every five years or a decade.

With the advancement of Information and Communication Technology (ICT), the positioning or context of users are relatively easy to collect. Some modalities are available, including GPS location data, Bluetooth or WiFi-based indoor localization, and MPD (Mobile Positioning Data) from communication provider. While the earlier requires specific devices (e.g., GPS-enabled devices and Bluetooth beacons [4], [5]), the latter is more ubiquitous and has a wider coverage. The whole set of mobile phones registered with the operator networks are available to track, thanks to the defined location of cellular base stations.

The present research explores mobile location data gathered from a telecommunication provider in Indonesia. Based on triangulation-based localization, the positions of users are estimated. We thus extract the mobility of people based on their position on Java island, one of the most populated islands in the world[4]. We finally elaborate the impact of the large-scale social restriction on peoples' weekly movement

[1]https://covid19.apple.com/mobility
[2]https://visualization.covid19mobility.org/
[3]https://www.google.com/covid19/mobility/
[4]https://finance.detik.com/berita-ekonomi-bisnis/d-5343868/sensus-penduduk-2020-selesai-begini-sebaran-masyarakat-ri-terbaru

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and *'libur lebaran'*, the homecoming trips due to Muslim Holy Days. The contribution of this paper are insights, based on the actual mobility data, on how the government decision affects the mobility of people in Indonesia.

The paper is organized as follows. Section II overviews the related work. We thus describe our methodology in Section III. In Section IV, we present and discuss our findings. Finally, Section V summarizes the findings presented in the paper.

II. Related Work

Mobility pattern analysis can be derived both through surveys and by utilizing some technologies or available data sources. The earlier is a traditional approach, where researchers or government authorities needs to conduct surveys to get information about people travel pattern on some previous days. The latter is considered to be a more sophisticated and technological advance, that is, utilizing vehicle detection technology or telecommunication-related dataset. Caceres et al. have compared mobility inference from both ways [6]. The authors find that even though mobile technology may raise questions for short trips ($< 2.5km$), this data type may indicate many common features with much less effort compared to labor-worked traditional surveys.

In this COVID-19 pandemic, many researchers have worked on mobile phone data to provide guidance and evaluation regarding COVID-19 response. Grantz et al. investigate information possibly gathered using mobile phone data and their possible biases [7]. The information includes CDR (Call data records), GPS location data, Bluetooth data, and opt-in contact tracing application data. These sources offer different levels of information granularity and in turn, different data biases and privacy aspects. While CDR or mobile telecommunication data cannot provide a very accurate location and cannot distinguish COVID transmission risks, this modality may be sufficient in analyzing people's mobility in a larger coverage. Furthermore, this source is readily available as this data is collected by default by telecommunication operators.

By using mobile device location, Warren and Skillman investigate the effect of mobility restriction to daily movement. The location data are anonymized or de-identified to avoid individual identification that leads to privacy violation [8]. According to their observation, mobility reductions happened in the US and some other countries, including Indonesia. However, the authors investigate mobility changes in more detail only in the US (up to county-level). More detail investigation play an important role in addressing the issue of disease transmission due to physical contact and mobility.

To investigate people's mobility, several metrics have been proposed and used. Warren and Skillman use Haversine distance / great circle, bounding box mobility, and convex hull mobility [8]. Pan et al. propose Social Distancing Index (SDI) to observe mobility changes during COVID-19 pandemic [9]. Pan et al investigate the impact of mandatory order to work at home and social distancing practicing to the strengthened practices. The states with more cases are more aggressive (active and timely responsive) than those with fewer cases. OD matrix has been used based on Facebook mobility data to map risk of COVID-19 outbreaks in Australia [10] Four people mobility features were used in [11], consisting of average distance traveled, stay-at-home rates, number of rides, and which Ward was moved to. Metrics they use are the number of trips, OD matrices, and traveled distance [6].

Studies of mobility in Indonesia commonly utilize open mobility reports generated by big companies, such as Google and Apple [12], [13]. In the datasets, people mobility is clustered in some particular areas, such as in public transit transportation, recreation, office, and residential areas. Nugroho et al. report that there is a correlation between daily COVID-19 new cases with people mobility in bus and train station (0.33), retail and recreation (0.25), and traditional market groceries and pharmacies (0.24) between March 21, 2020, and March 3, 2021 [13]. Auliya et al. investigate the impact of the social restriction on people's movement and how their mobility affects the disease transmission rate [12]. Similar to Nugroho et al., the authors observe a correlation between mobility trends and daily new cases and report that the new cases are 8 days behind the mobility trend according to the modified Pearson's correlation formula. Additionally, they report that people's mobility in Jakarta and Indonesia (nationwide) was significantly decreased in March 2020 and started to run normally in mid-May 2020 (by driving activity). This paper attempts to extract Indonesian people's mobility from the available mobile telecommunication data. The analysis will cover second-order administrative division (i.e., provinces in Indonesia) in Java island and the aggregation of them.

III. Methodology

A. Data Collection

We use the location data of hashed cell phones provided by a telecommunication provider in Indonesia. The location of users are gathered based on a standard triangulation algorithm with the accuracy about 200- to 300-meters [14]–[16]. This process is done in provider side. The data is then transferred periodically by the telecommunication provider to our landing server. Every day, we receive a massive amount of data (300 Giga Bytes) in the landing server. To deal with large data size, we forward the data to two different servers: Hadoop-based data lake, which stores complete daily data, and Elasticsearch database, which stores only newer data in a certain period. The latter allows faster queries for data analysis.

We randomly select 1000 people located in five provinces in Indonesia (DKI Jakarta, West Java, Central Java, DI Yogyakarta, and East Java) during three following months, March-May 2020. This is the period when the Indonesian government strengthened measures in limiting people's mobility due to COVID-19.

B. Data Processing

The collected data have high volumes of mobile positioning. We need to get the data that we will process. In this work, we filter user data in five provinces in Java island, Indonesia. Next, we do data cleaning to eliminate duplication

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TABLE I
THE EXTRACTED DATA FIELDS

Fields	Remarks
Time Stamp	Date and time format where the transaction is recorded
MSISDN	Unique number represents cell phone number which has been hashed by data provider.
Latitude	Measurement of distance north or south from the equator
Longitude	The measurement of distance east or west from prime meridian

and repair incomplete or inconsistent data. Finally, we extract important fields, such as timestamp, msisdn, and coordinates as shown in Table I. These fields are then analyzed to see mobility pattern. The mobility analytic requires additional information that indicates in which administrative area (i.e., province) a msisdn is located. As the raw data only contains coordinates (i.e., latitude and longitude), we map the coordinates to provinces by querying them based on polygon of the provinces. We utilize generic Convex-hull algorithm for query purposes.

C. Metrics

In order to evaluate people mobility, we accommodate several metrics:

1) Number of visits: The number of visits is counted based on individual movement paths. We count a number of trajectories to certain places. Visitation to the same places could be counted as multiple time of visits, for example when an individual visits the same place on another occasion. As the obtained coordinate location is an approximation based on signal triangulation, the coordinate is not as accurate as the location obtained based on GPS. Therefore, we cluster a group of nearby coordinates as a relevant location to compromise inaccuracy in location determination.

2) Traveled distance: Traveled distance represents historical movement on an observed day. The movement starts from a home location that is determined based on a dominant position in the evening. We calculate the traveled distance between two locations based on sorted timestamps for each unique MSSIDN. Traveled distance is estimated using latitude and longitude based on haversine formula [17]. This computation represents distances on the surface of a sphere. We accumulate the distance of each individual during the day and finally calculate the averaged traveled distance during the observation period.

3) Origin-Destination Matrices: Origin-Destination matrix represents the number of trips that were made between two locations in certain periods [7]. We observe trips between two administrative areas. We consider the administrative area in the level of provinces as we want to summarize people mobility in a large area, within a province, or between two provinces. The trips consist of trip production and trip attraction. Trip production is the number of outgoing trips from an administrative area, while trip attraction is the number of incoming trips to an administrative area from any other region.

IV. RESULTS AND DISCUSSIONS

Based on the mobile positioning data, we analyze the mobility of people in five provinces on Java island, Indonesia. People's mobility is indicated by several metrics, such as the number of visits, traveled distance, and Origin-Destination matrices.

A. Number of visits and traveled distance

Figures 1-5 show people mobility, i.e., represented by average traveled distance and number of visits, in five provinces in Indonesia, namely, D.I. Yogyakarta, DKI Jakarta, Central Java, West Java, and East Java. In general, the mobility pattern is similar, but the magnitude is slightly different. One can see that in the first three weeks (9-29 March 2020), the average travel distance is high in all provinces, yet the average visit remains low. Yogyakarta province reaches the highest average traveled distance among the other provinces. Figure 1 shows that during the period, people traveled almost ten kilometers per week with average visit of 300 places in Yogyakarta.

The Indonesian government decided to enforce online schooling and 'working from home (WFH)' for non-essential public workers on 24 March 2020 [18]–[20]. As a response, several local governments regulate social restrictions on various scales, such as in DKI Jakarta, a large-scale social restriction was started on 10 April 2020. These policies significantly reduced the average traveled distance in the fourth week (29 March - 4 April) and reached the lowest in the fifth week (5 - 11 April) for all observed provinces. While the traveled distance decreased, however, the average number of visits increased (orange line in the figures). Until the traveled distance went up again in the sixth week (12 - 18 April), the number of visits continued to grow. The number of visits reached a peak in the eighth week of observation in all provinces, reaching of approximately 960 and 2760 average visits in Central Java and Yogyakarta, respectively. These numbers are four times and nine times higher than the number of visits in the first weeks of observation when schools, universities, and offices remained open, respectively. This phenomenon (i.e., the short traveled distance while the number of visits becomes higher) indicates that people move around more frequently. A possible explanation is that while most schools and offices were closed and most activities were done online, public spaces remained open, such as in Yogyakarta [21] and Semarang, Central Java [22].

The government strengthened the measure to prevent people traveled back to their hometown due to Libur Lebaran in the ninth week (beginning of May). It made the number of visits under control again, below 30 visits in all provinces. On 24-25 May, there was a small spike of the number of visits due to Idul Fitri.

We can infer a general pattern that on the 5th and 14th weeks the average distance was at a low level followed by significant changes in the 6th and 15th week in all provinces. The growth is shown in Figure 6. The government restriction on mobility resulted in a significant reduction on three weeks after its announcement until the 5th week of observation

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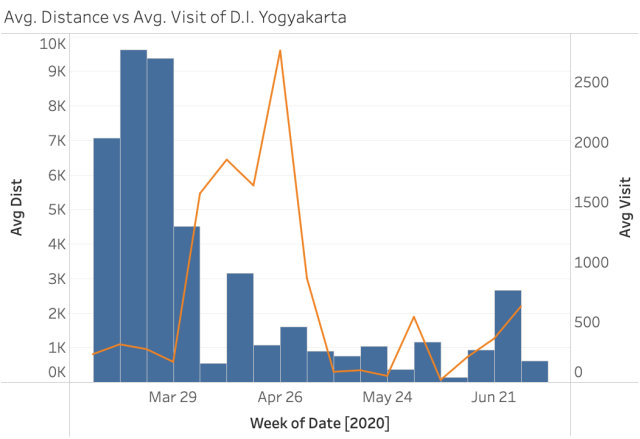

Fig. 1. Mobility pattern in Yogyakarta

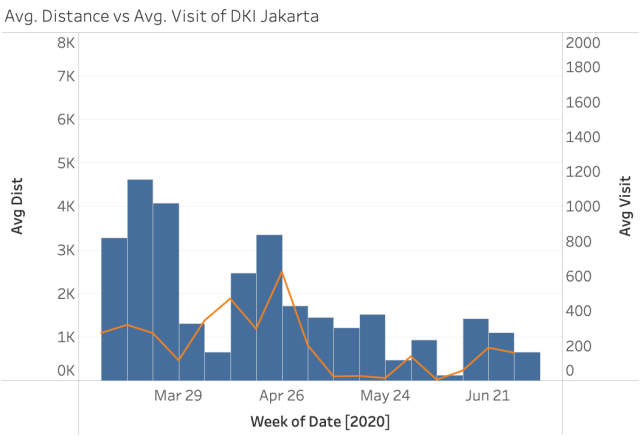

Fig. 2. Mobility pattern in DKI Jakarta

(i.e., 11 April 2020). In the following week (the 6th week on 12-18 April 2020), there was 278.4% average distance growth. From the observation of individual provinces, the lowest average distance was in the 14th week (7-13 June 2020). This was followed by a significant increase in the 15th week, reaching 763.2%. A likely explanation is that

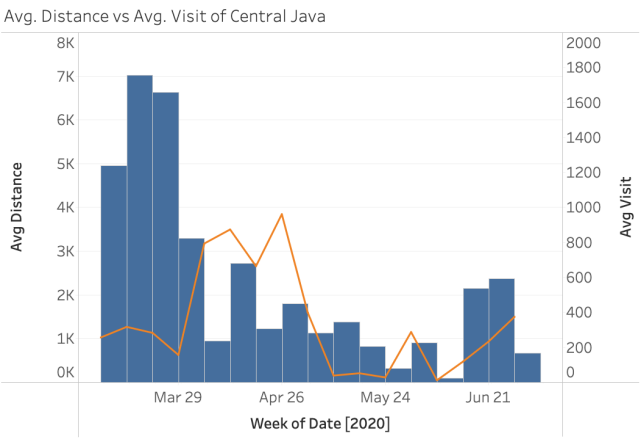

Fig. 3. Mobility pattern in Central Java province

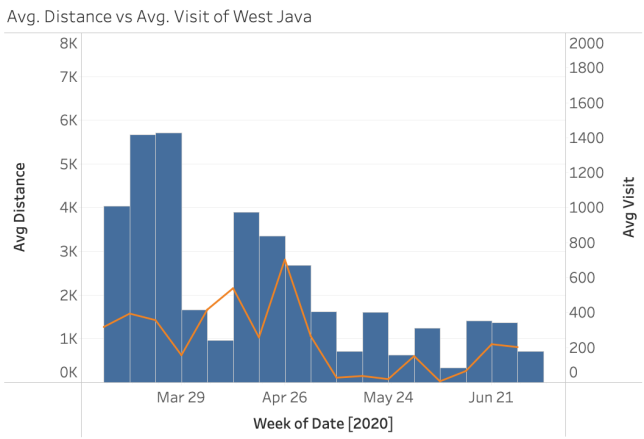

Fig. 4. Mobility pattern in West Java province

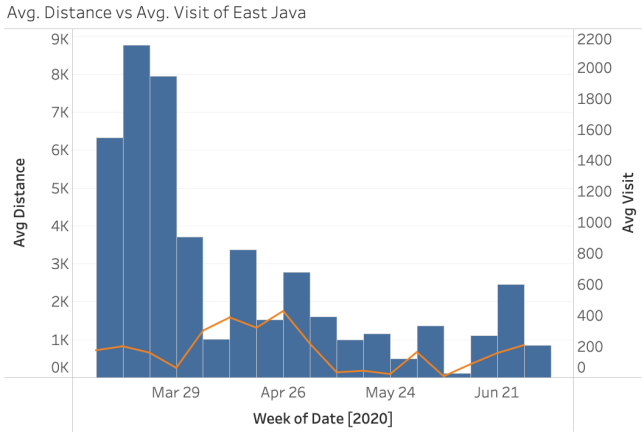

Fig. 5. Mobility pattern in East Java province

government starts to ease restrictions, such as allowing 80 shopping malls to operate starting from 15 June 2020 [23], [24].

Despite the significant spikes, the accumulative distance growth from the first week shows a consistent reduction in the traveled distance as shown in Figure 7. The government restriction in the early phase resulted in traveled distance reduction reaching -83.9% (cumulative) since the first week of observation. The lowest cumulative traveled distance reduction was -96.8%. It happened before the government allowed businesses to operate in the transition phase in the 15th week.

B. Origin-Destination matrix

The trips that have been made during the observation period are summarized in Figure 8. The Origin-Destination (OD) matrix represents the number of trips between two administrative areas (e.g., provinces), We can see that the most frequent trips are within Jawa Tengah and between DIY and Jawa Tengah, reaching almost 2.5 and 5.4 million trips, respectively. On the other hand, the number of trips within DKI Jakarta province is only 518,627 trips. A likely explanation is that large-scale social restriction was applied in DKI Jakarta, and there is no further restriction in Jawa

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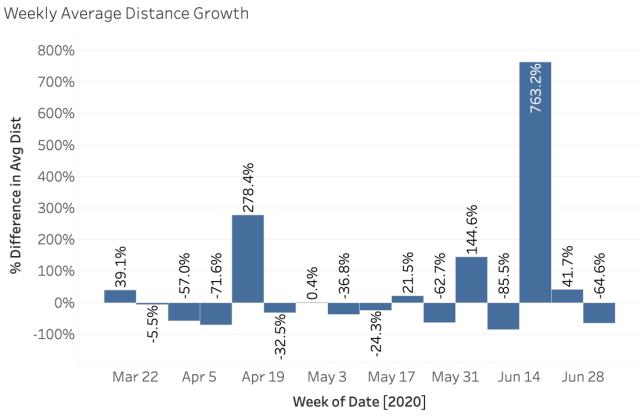

Fig. 6. distance growth

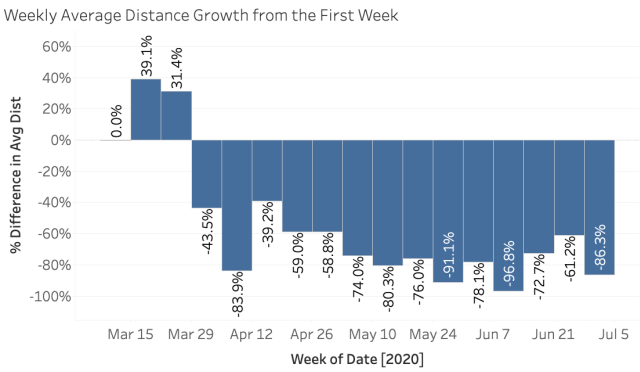

Fig. 7. distance growth from 1st week

Tengah and DIY. Another important finding is that during large-scale social restriction in DKI Jakarta, the number of trips to other provinces is very limited, except to Jawa Barat province where satellite cities of the national capital Jakarta reside (e.g., Bogor, Depok, Tangerang, and Bekasi).

Trip attraction and trip production have a similar pattern and magnitude. It indicates that the number of people coming in to and going out from a province is equal. In other words, people travel for commuting trips and they are expected to come back on the same day. There is a peak of the number of trips in Week 18 (26 April - 2 May 2020) in Jawa Tengah and DI Yogyakarta. This peak could be due to a part of a tradition of welcoming the holy month of Ramadhan (i.e., *'Nyadran'* or *'Ziarah kubur'*) [25].

OD Matriks Total

	Destination				
Origin	DAERAH ISTIMEWA Y..	DKI JAKARTA	JAWA BARAT	JAWA TENGAH	JAWA TIMUR
DAERAH ISTIMEWA YOGY..	2.591.302	2		5.477.822	10
DKI JAKARTA	1	518.627	872.119	6	7
JAWA BARAT	1	871.935	2.687.115	55.162	
JAWA TENGAH	5.477.828	4	55.186	5.444.091	157.460
JAWA TIMUR	11	5	3	157.444	1.512.544

Fig. 8. OD matrix

V. Conclusion and Future Work

Based on the investigation on the Mobile Positioning Data in Indonesia, some insights are inferred.

a) Mobile Positioning Data shows people mobility patterns w.r.t. the avg. traveled distance and number of visits: In the normal condition before the social restriction, the average traveled distance is relatively high (i.e., more than 5 kilometers) and the average number of visits is low (i.e., fewer than 400). The social restriction reduces the traveled distance, but the number of visits spikes up to 9 times (i.e., reaching 2760 visits in Yogyakarta). It indicates that people do not travel a long distance, but they are still having trips with shorter destinations.

b) The government intervention on mobility trips was successful: The weekly mobility trend (i.e., towards the traveled distance) shows a declining direction. It indicates that people's mobility growth has been suppressed by implementing social restrictions. Overall, the accumulative traveled distance declines more than 8.3 times compared to the normal condition. Especially in Ramadhan month, the government intervention on social restriction has reduced the number of visits, reaching lower than 30 visits. We can see from the OD matrix that the trips during the period are local journeys, instead of coming home trips for *'lebaran'*. There are more trips between neighboring provinces (e.g., Jawa Tengah and DIY), but it is not during the *'Libur Lebaran'* period.

c) Relaxation on Mobility Restriction may significantly affect travel distance growth : While the trend of traveled distance is declining, there is a potency that it can rocket up to 7.6 times in a week when the government decides to relax the social restriction. Hence, the government needs to take a careful and thorough decision when allowing mobility to become less strict.

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References

[1] Y. Gao, J. Liu, Y. Xu, L. Mu, and Y. Liu, "A Spatiotemporal Constraint Non-Negative Matrix Factorization Model to Discover Intra-Urban Mobility Patterns from Taxi Trips," *Sustainability*, vol. 11, no. 15, p. 4214, Jan. 2019, number: 15 Publisher: Multidisciplinary Digital Publishing Institute. [Online]. Available: https://www.mdpi.com/2071-1050/11/15/4214

[2] C. O. Buckee, S. Balsari, J. Chan, M. Crosas, F. Dominici, U. Gasser, Y. H. Grad, B. Grenfell, M. E. Halloran, M. U. G. Kraemer, M. Lipsitch, C. J. E. Metcalf, L. A. Meyers, T. A. Perkins, M. Santillana, S. V. Scarpino, C. Viboud, A. Wesolowski, and A. Schroeder, "Aggregated mobility data could help fight COVID-19," *Science*, vol. 368, no. 6487, pp. 145–146, Apr. 2020. [Online]. Available: https://www.science.org/doi/10.1126/science.abb8021

[3] N. McGuckin and A. Fucci, "Summary of travel trends," *National Household Travel Survey, Washington DC USA*, 2017.

[4] A. R. Pratama, A. Lazovik, and M. Aiello, "Office multi-occupancy detection using ble beacons and power meters," in *2019 IEEE 10th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*. IEEE, 2019, pp. 0440–0448.

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[5] S. Boonsriwai and A. Apavatjrut, "Indoor wifi localization on mobile devices," in *2013 10th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology*, 2013, pp. 1–5.

[6] N. Caceres, L. M. Romero, and F. G. Benitez, "Exploring strengths and weaknesses of mobility inference from mobile phone data vs. travel surveys," *Transportmetrica A: Transport Science*, vol. 16, no. 3, pp. 574–601, Jan. 2020. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/23249935.2020.1720857

[7] K. H. Grantz, H. R. Meredith, D. A. T. Cummings, C. J. E. Metcalf, B. T. Grenfell, J. R. Giles, S. Mehta, S. Solomon, A. Labrique, N. Kishore, C. O. Buckee, and A. Wesolowski, "The use of mobile phone data to inform analysis of COVID-19 pandemic epidemiology," *Nature Communications*, vol. 11, no. 1, p. 4961, Dec. 2020. [Online]. Available: https://www.nature.com/articles/s41467-020-18190-5

[8] M. S. Warren and S. W. Skillman, "Mobility Changes in Response to COVID-19," *arXiv:2003.14228 [cs]*, Mar. 2020, arXiv: 2003.14228. [Online]. Available: http://arxiv.org/abs/2003.14228

[9] Y. Pan, A. Darzi, A. Kabiri, G. Zhao, W. Luo, C. Xiong, and L. Zhang, "Quantifying human mobility behavior changes in response to non-pharmaceutical interventions during the COVID-19 outbreak in the United States," *arXiv:2005.01224 [cs]*, May 2020, arXiv: 2005.01224. [Online]. Available: http://arxiv.org/abs/2005.01224

[10] C. Zachreson, L. Mitchell, M. J. Lydeamore, N. Rebuli, M. Tomko, and N. Geard, "Risk mapping for COVID-19 outbreaks in Australia using mobility data," *Journal of The Royal Society Interface*, vol. 18, no. 174, p. 20200657, Jan. 2021. [Online]. Available: https://royalsocietypublishing.org/doi/10.1098/rsif.2020.0657

[11] S. Isoda, S. Kawanaka, Y. Matsuda, H. Suwa, and K. Yasumoto, "How much does human mobility behavior affect the COVID-19 infection spread?: poster abstract," in *Proceedings of the 18th Conference on Embedded Networked Sensor Systems*. Virtual Event Japan: ACM, Nov. 2020, pp. 766–767. [Online]. Available: https://dl.acm.org/doi/10.1145/3384419.3430593

[12] S. F. Auliya and N. Wulandari, "The Impact of Mobility Patterns on the Spread of the COVID-19 in Indonesia," *Journal of Information Systems Engineering and Business Intelligence*, vol. 7, no. 1, pp. 31–41, Apr. 2021, number: 1. [Online]. Available: https://e-journal.unair.ac.id/JISEBI/article/view/25241

[13] L. E. Nugroho and A. Z. Rakhman, "Mobilitas Manusia dan Tingkat Penyebaran Covid-19: Sebuah Analisis Kuantitatif," *Jurnal Nasional Teknik Elektro dan Teknologi Informasi*, vol. 10, no. 2, pp. 124–130, May 2021, number: 2. [Online]. Available: https://jurnal.ugm.ac.id/v3/JNTETI/article/view/1519

[14] J. Yang, A. Varshavsky, H. Liu, Y. Chen, and M. Gruteser, "Accuracy characterization of cell tower localization," in *Proceedings of the 12th ACM International Conference on Ubiquitous Computing*, ser. UbiComp '10. New York, NY, USA: Association for Computing Machinery, 2010, p. 223–226. [Online]. Available: https://doi.org/10.1145/1864349.1864384

[15] S. Jiang, G. A. Fiore, Y. Yang, J. Ferreira Jr, E. Frazzoli, and M. C. González, "A review of urban computing for mobile phone traces: current methods, challenges and opportunities," in *Proceedings of the 2nd ACM SIGKDD international workshop on Urban Computing*, 2013, pp. 1–9.

[16] Y. Zhao, "Standardization of mobile phone positioning for 3g systems," *IEEE Communications Magazine*, vol. 40, no. 7, pp. 108–116, 2002.

[17] N. R. Chopde and M. Nichat, "Landmark based shortest path detection by using a* and haversine formula," *International Journal of Innovative Research in Computer and Communication Engineering*, vol. 1, no. 2, pp. 298–302, 2013.

[18] "SE Mendikbud: Pelaksanaan Kebijakan Pendidikan dalam Masa Darurat Penyebaran Covid-19," Mar. 2020, section: Pengumuman. [Online]. Available: https://www.kemdikbud.go.id/main/blog/2020/03/se-mendikbud-pelaksanaan-kebijakan-pendidikan-dalam-masa-darurat-penyebaran-covid19

[19] F. Ludiana, "Pencegahan Penyebaran Virus Covid-19 dengan Kerja di Rumah bagi ASN." [Online]. Available: https://menpan.go.id/site/berita-terkini/pencegahan-penyebaran-virus-covid-19-dengan-kerja-di-rumah-bagi-asn

[20] Sihono, "UNNES Terapkan Kebijakan WFH." [Online]. Available: https://unnes.ac.id/berita/unnes-terapkan-kebijakan-wfh.html

[21] "Kasus Positif Corona di DIY Melonjak, Ini Kata Sultan HB X." [Online]. Available: https://news.detik.com/berita-jawa-tengah/d-4996719/kasus-positif-corona-di-diy-melonjak-ini-kata-sultan-hb-x

[22] "Menyoal Pembatasan Kegiatan Masyarakat di Semarang, Apa Bedanya dengan PSBB? Halaman all - Kompas.com." [Online]. Available: https://regional.kompas.com/read/2020/04/28/11330061/menyoal-pembatasan-kegiatan-masyarakat-di-semarang-apa-bedanya-dengan-psbb-?page=all

[23] "Shopping malls, M Bloc Space among businesses reopening on Monday - Lifestyle - The Jakarta Post." [Online]. Available: https://www.thejakartapost.com/life/2020/06/15/shopping-malls-m-bloc-space-among-businesses-reopening-on-monday.html

[24] F. M. Ramadhan, "80 Mal Jakarta Dibuka saat PSBB Transisi, Apa yang Masih Tutup?" Jun. 2020, section: Metro. [Online]. Available: https://grafis.tempo.co/read/2106/80-mal-jakarta-dibuka-saat-psbb-transisi-apa-yang-masih-tutup

[25] T. J. Post, "Nyadran, a unique tradition of welcoming Ramadhan." [Online]. Available: https://www.thejakartapost.com/life/2017/05/16/nyadran-a-unique-tradition-of-welcoming-ramadhan.html

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